80 REVIEWS

"We once received a tape with a program on one side and Barry Manilow's Greatest Hits on the other."

Acu-Data Tape Digitizer
Alphanetics Mfg.
Forestville, CA
$54.95

by Chris Brown
***80* Staff**

Ah CLOAD, a much maligned mode. That the process works at all is testimony to the flexibility of the electronic devices involved. Cassette recorders were never really meant to handle data, even at slow rates, and microprocessors wince at the thought of their internal timing being upset by wildly gyrating cassette tape transport mechanisms.

Nevertheless, if you persevere, you can often actually load a program from tape into a computer via a cassette recorder. It may take some time, but it can be done. If your time (and patience) is at a premium however, you should consider acquiring an Acu-Data from Alphanetics.

Data Pulses are Shaped

The Acu-Data is designed to facilitate CLOAD and tape duplication. In essence, it is a combination filter, rectifier and pulse shaping device. Placed at the recorder output, it conditions the tape data. Hum and other spurious noise is filtered out. Data pulses are then rectified and shaped to insure that proper pulse amplitude and timing is achieved before the computer ever sees the pulse train.

An Acu-Data has been in use at the *80 Microcomputing* editorial office for six months. Quite simply, it works.

Program tapes of questionable quality often accompany article submissions to the magazine, and loading these gems can be a real epic. We once received a tape with a program on one side and "Barry Manilow's Greatest Hits" on the other. The Acu-Data rarely encounters a tape it can't handle (including Barry Manilow's).

The unit runs on 110VAC so batteries are not necessary. Other features include an LED to indicate the presence and amplitude level of data, a polarity switch that allows the user to select either positive or negative going data pulses (to compensate for differences in the head and audio circuits or various cassette recorders) and a copying digital output jack that allows duping of processed program tapes.

The model we have has an optional switch that provides computer control of the recorder. This option, especially handy when sequentially loading data in long programs, costs an extra five bucks.

The most immediate benefit of an Acu-Data is that it ends your worries about recorder volume settings. The unit produces the proper output level under widely varying input conditions: No more fooling with your kid's Dick Tracy Wrist Radio and the recorder volume control to get a good load.

The Acu-Data has proved to be a miracle worker when it comes to salvaging marginal tapes.

In addition, lousy originals can generate good copies when the copying digital output is used for duplicating.

There are limits to the capabilities of the Acu-Data. Generally though, if you could not quite load a tape through the Acu-Data, you would not have gotten close without it.

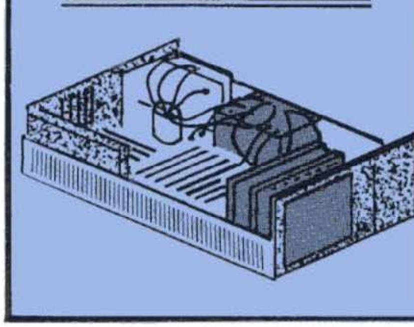
The unit is ruggedly built, enclosed in a shielded metal case. The version we have shows signs of last minute fixes on the clad side of the PC board. The manufacturer assures us that they are not present on current production models.

Always On Line

The Acu-Data is on line whenever it is plugged into a 110VAC outlet. No provision for removing AC from the primary of the line transformer is made and, consequently, the transformer is forced to dissipate a respectable amount of hysteresis generated heat. How this constant heat dissipation will effect its service life is a good question. So far, it has not affected the performance of our unit.

The comprehensive user's manual is clearly written and includes many hints on using tapes.

HARDWARE



A schematic and parts list was not available, but we did see the warranty. It is impressive. Each Acu-Data comes with a ten-day, money back guarantee. In addition to a 90 day, unconditional warranty on parts and labor, a flat $15 maximum repair fee is guaranteed on any problems encountered within the first 12 months.

Guarantees like this are rare in the microcomputer industry and certainly inspire confidence. Three cheers for Alphanetics.

Alphanetics has a winner in the Acu-Data. For tape oriented computerists who prefer to think in terms of black boxes and have no desire for breadboard projects, the Acu-Data is a worthwhile investment at $54.95. ■

TC-8 Cassette System
JPC Products
Albuquerque, NM
Kit: $70
Assembled: $100

by Carl A. Kollar

I guess I don't have to tell any TRS-80 owners how frustrating the cassette system that comes with the computer can be. Even with the factory mod that's available, the annoyance of loading and checking programs becomes just barely tolerable.

If you're like me, after you've just plunked down a chunk of money for a Level II 16K machine, "you ain't got nuttin left" for even one disk drive at 500 bucks apiece. So you suffer.

A reasonable alternative is the Exatron Stringy Floppy (ESF). This will cost you about 250 bucks and totally eliminates your loading and saving problems, automatically and fast. I've had one of these for about six months and love it!

But, if the price is still too steep, have I got a device for you!

The Device

The February 1980 issue of *Microcomputing* had an ad that intrigued the hell out of me. It was for a high-speed cassette system by JPC Products acclaimed as a "poor man's floppy." It made all sorts of seemingly ridiculous claims such as "loads five times faster," "stores 50,000 bytes on a 10-minute cassette," "less than one bad load in a million bytes with the volume control anywhere between one and eight."

All this for a measly 70 bucks? How could this be? A call to Albuquerque answered a few questions: Yes, it had its own power supply,

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and, it stored programs five times faster because it utilized higher density data. The computer outputs the information at a higher rate out of the rear keyboard connector.

The ad had even claimed anyone could build it even if you have never soldered before. JPC would make it work, if you couldn't—for free. I was sold. I placed my order, and it arrived about two months later (parts shortage).

I work in electronics, so I found the unit exceptionally easy to build. It took about an hour. The manual is superb. (That's better than great.) It was clear, concise and exact with no ambiguities. Important parts placements are stressed (polarity markings on electrolytics, bands on diodes, etc.).

JPC was right! With these instructions, you couldn't go wrong. The board quality is excellent. It is double-sided and parts locations are clearly marked on the component side of the board. There are no jumper wires to install. JPC utilizes PC traces and plated-through holes for connections to traces on the other side of the board.

Also, there are absolutely no adjustments or settings to bother with.

The documentation is a sheaf of 8½ x 11 papers stapled together. It is written in the nicest format I've seen in a while. Each command and/or subject is covered on its own sheet in large type. All explanations are in easy to read English—not computerese.

Commands and Features

SAVE"filename": Saves your BASIC program on cassette.
LOAD: Reads the next BASIC program from the cassette.
LOAD"filename": Searches for and loads the specified file from cassette.
LOAD? and LOAD? "filename": Reads file from cassette, and compares contents to memory.
LOADN: Prints a list of all the programs on a cassette, until interrupted by the "break" key.
LOADN"filename": Same as above except the tape will stop at the end of the program named.
KILL: Removes the file manager program from memory so that the extra memory can be used by large programs.
RSET: Allows the operator to rewind and position the tape on tape recorders that have these functions tied to the motor control jack.
RUN"filename": TC-8 searches for a specified program and runs it immediately.
PUT"filename": Same as SAVE"filename", except it is for use with system tapes.
GET: Same as LOAD, except it is for use with system tapes.
GET"filename": Same as LOAD"filename", except it is for use with system tapes.
GET? and GET? "filename": Same as LOAD? and LOAD?"filename", except it is for use with system tapes.
GETN and GETN"filename": Same as LOADN and LOADN"filename", except it is for use with system tapes.
OPEN: Required before cassette input or output of a data file can be attempted.
CLOSE: Required to end a cassette data file.
PRINT#: Allows numerical or string data to be output to a cassette file.
INPUT#: Allows numerical or string data to be input from a cassette file.

I haven't counted them, so I don't know about the "one load in a million bytes" claim, but my son, Anthony (age 11), loaded about 30 of his programs from his Radio Shack format tape to a new TC-8 format tape. He's run them all and found no bad loads.

Unlike the standard tape system, you can position your tape anywhere before the program you want and not have to look for a blank spot between programs. The TC-8 patiently waits for the program you want and then starts loading without getting confused by the portion of the previous program you just fed it.

Try that on your regular cassette system; you'll wear out the reset button. ■

MAYDAY +S
Uninterruptable Power Supply
Sun-Research, Inc.
New Durham, NH
$325

by Chris Brown

Mayday, the international signal of distress, is not likely to be the first word uttered by a computerist when the power fails in the middle of a lengthy program. Personally, I can think of many more satisfying expletives, but Mayday may be more appropriate. It is the trade name for an uninterruptable power supply, designed to end power line problems.

Emergency Power

Mayday is a fail-safe device which, in the event of a drop in line voltage, provides emergency power to a microcomputer system. Once on emergency power, you can terminate the system in an orderly fashion with no program crash or loss of data. Emergency operation time varies with the Mayday model and the size of your system, that is, how many disk drives you own. The Mayday +S can generate up to 30 minutes of emergency power.

Sun-Research, Inc. is a new manufacturing and marketing venture launched by Phase-R Corporation, a diversified New Durham electronics firm. Among its products are laser devices manufactured for the medical field and the government.

The original Mayday unit was created to meet Phase-R's need for a reliable, isolated power source for the office TRS-80. The company's rural location resulted in frequent power outages, and the TRS-80's close proximity to heavy machinery made it difficult for it to run without glitches. After a few monthly payroll records were lost, the Mayday was born.

Not only does the unit provide instant backup power, it also isolates the computer from spikes and transient voltage surges on the AC line.

The Mayday uses a modified, 120 Hz, square wave generator as a DC to AC converter. When power fails, this generator supplies power from a 12-volt battery to the computer. A specially designed isolation transformer allows the Mayday to maintain, plus or minus, five percent required computer power during switch-over to internal power. Plus or minus one-half percent is maintained thereafter. Switch-over time is on the order of five milliseconds, so no loss of memory occurs.

A 12-volt automotive battery (available separately) is used in conjunction with the Mayday and is enclosed in a high impact plastic case. A built-in trickle charger keeps the battery voltage within accepted limits.

The computer monitor, interface, keyboard and peripherals plug into a bank of outlets on Mayday's front panel. The entire system is controlled by one circuit breaker, while an idiot light indicates system status. A very convenient layout.

Line Surge Protector

Most versions of the Mayday incorporate an MDS line surge protector. If your computer operates in an electrically noisy environment, this option is a must.

If its operation in our office is any indication, Sun-Research has met its design goals with the Mayday. The 80 editorial offices share building space with a printshop and darkroom, and the AC line that our computer runs on is subject to large voltage swings and noise. The constant shrinking of our video as presses came on line was a real source of worry.

Once the Mayday +S was installed, the shrinking video was eliminated. Now, the occasional excursions of the line below normal levels result in a smooth transition to emergency power rather than a catastrophic program crash.

One drawback of our Mayday +S is its inability to power the 60 Hz AC fan motors used in our disk unit and printer. This can be overcome by wiring these motors directly to an external AC source. To alleviate this problem, Sun-Research has developed a Mayday unit that supplies a 60 Hz sine wave source of voltage. This new Mayday is suitable for use with Model II machines.

As the air conditioning season approaches and brown-outs become more frequent in metropolitan areas, the computer chaos caused by erratic AC lines will increase. A Mayday unit can be your first line of defense against system glitches. Now, if you can only get your TRS-DOS squared away . . . . ■